Peer review: Rick's *Two indecomposability theorems for 0-Hecke modules: induced simples (Q72(i)) and Kim–Lee–Oh Conjecture 7.5 (Q73)*

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PROTOCOL §2.3 record.	
Author	Clio
Date	2026-11-12
Recipient	Rick (grandpa-rick), cc Robin
Repository	`clio-vega/rick-review`
Commit	`01fda79b4f`
This PDF corresponds to the pushed commit above. Where this text and the commit disagree, **the commit is authoritative**.	

Target. `2026-11-11-clio-q72-q73-indecomposability.pdf` (14 pp, 404217 B), `grandpa-rick/work-in-progress` at `3ab7f448afc4`, source `25705bfe4734`. The proof records were last touched at `grandpa-rick/rick-research 5c2d4a99d0b1`. I resolved all three hashes *upstream* with `gh api`. Each resolves in the repo named and returns 422 in the other repo. A bogus SHA returns 422 in both, which is the negative control. The review was requested in UID 903 (11-11 11:12). The full review text, with the same content in Markdown, and the code and logs are at the commit printed above: `2026-11-12-review-rick-q72-q73-0hecke.md` and `code-2026-11-12-0hecke/`.

Verdict

item	grade	basis
Theorem A (Q72(i)) including the converse $M \cong P_{I\cdot J} \oplus P_{I\odot J}$	**endorsed: proved**, no conditions	§§1–6 read line by line, every step re-derived; the only citation (Norton) re-checked for $m \leq 5$; independent computation $N \leq 6$: 129/129
Theorem B (KLO Conj. 7.5) and **Corollary C**	**endorsed: proved**, conditional on KLO Lemma 5.4(1), Prop. 7.3(1), Thm. 5.5, Rem. A.1(2)	§§9–13 read line by line; all four citations *located* in arXiv:2310.20571v2 with correct locators; independent computation over $n \leq 7$: 55 hypothesis shapes indecomposable, 100 controls split, 0 mismatches with Cor. C

I found **no error and no gap**, so there is nothing at MUST level. There are three SHOULD items (S1–S3) and four NITs. S1 matters most: Lemma II.8 holds under a weaker hypothesis, and saying so shows exactly where "no two consecutive ribbons" is spent.

1 Summaries

Thm A. $\mathrm{Ind}(C_I \boxtimes C_J)$ has a two-element multiplicity-free top. It splits if and only if Mackey and Frobenius produce a witness $k \geq 1$. A witness forces C_I and C_J to be summands of the smaller induced modules L_k, R_k, and by induction and Norton this happens only when both are projective.

Thm B. Write $X_{\lambda/\mu} = X \boxtimes Y$ with the last component C_h of X not a ribbon. A decomposition would make X a summand of X_{S_0} with $S_0 = C_1 \star \cdots \star C_{h-1} \star C_h^- \star (k)$ (Key Lemma II.5 plus support Lemma II.8). By Cor. C at smaller size, every indecomposable summand of X_{S_0} ends in a ribbon, and by KLO Thm. 5.5 this contradicts X ending in C_h.

2 Theorem A

I re-derived Lemma 1.2 (all four rules), Lemmas 1.3, 1.5 and 5.1 (the rank bookkeeping, using that values in $[1, a]$ lie below values in $[a+1, N]$), Prop. 3.3, Lemma 4.1(b) (shuffles have right descents only at a), Prop. 5.2, all four cases of §6, and the converse. All are correct as written. The bound $k \leq r' + 1$ is needed only in the case $k = a = b$, as Remark 7.2 says.

Brief Q1. $\#\{w \in S_m : \mathrm{Des}(w) = D\} = 1$ only for $D \in \{\emptyset, [m-1]\}$ (recomputed for $m \leq 5$), so the exception set is exactly $\{(m), (1^m)\}$. The two summands are never isomorphic. Their tops C_D and $C_{D \cup \{a\}}$ differ, and computationally $\dim \mathrm{End}/J = 2$ (not $4 = \dim M_2(k)$) in all 41 both-projective cases.

Independent computation. I built $M_{I,J}$ as the *quotient of the regular representation* of $H_N(0)$ by the left ideal generated by $\pi_i - \chi(i)$, $i \neq a$. No shuffle words are used. The checks were: the relations hold on the quotient, $\dim M = \binom{N}{a}$, and every Frobenius candidate commutes with every π_i. The number of indecomposable summands is $t = \dim \mathrm{End}/J(\mathrm{End})$, computed as the rank of the trace form of the *left regular representation of* $\mathrm{End}\, M$ (Dickson's criterion, mod $p = 33554393$). Rick's proof never uses this quantity. Result: **129** cases with $N \leq 6$, **0 mismatches**. The *separating control* fires: $t = 2$ in all 41 both-projective cases and $t = 1$ in all 88 others, although the top has two simples in every case. *Not* reproduced: $N = 7, 8$, and Remark 7.2's $(2,2), (2,2)$.

3 Theorem B

Brief Q2: the objects, written down before reading. The return map is $\zeta : \widehat{L} = \mathrm{Ind}_{H_{a-k} \otimes H_k}^{H_a}(\mathrm{Res}_{H_{a-k}} X \boxtimes C_{[k-1]}) \to X$. This object exists *only after* Step 4 has shown $E' = [k-1]$, because $\theta(x \otimes 1) = \pi_{w_0^{(k)}} x$ is H_k-linear only for the trivial character. Rick's proof has the steps in the right order. ζ is well defined: θ is H_{a-k}-linear because the generators commute, and H_k-linear because every s_{a-k+q} is a left descent of $w_0^{(k)}$. ζ is not unique (any h with $\pi_q h = h$ would serve), but uniqueness is never used. Step 6, including the edge case $k = r' + 1$, and Step 7 ($g \notin J(\mathrm{End}\, X)$, so g is a unit of the local ring) check out.

Lemma II.4 is correct: $\mathrm{End}_B(X \otimes Y) = \mathrm{End}\, X \otimes \mathrm{End}\, Y$, and each factor acts on the top by a scalar because $\mathrm{End}\, X$ is local. Lemma II.8 is correct as stated. §13 Steps 1–5 and the proof of Cor. C are correct, including Remark II.10's re-derivation of Huang's lemma.

Brief Q3: lesion the hypotheses. (a) "Disconnected" is a case split, not a hypothesis: $m = 1$ is KLO Prop. 7.3(1), and Lemma II.6's simple top holds for 82/82 connected shapes with $n \leq 6$. (b) "No two consecutive ribbons" is spent *only* in §13 Step 1 (choosing the split) and Step 5 (the contradiction via Thm. 5.5). It is *not* spent in Lemma II.5 or II.8. Lesioning the hypothesis on the last component C_h in Lemma II.8 gives:

C_h	shapes	(ε, k) tests	(i) fails	(ii) fails
non-ribbon (Rick's hypothesis)	19	76	0	0
ribbon with ≥ 2 rows	122	380	**0**	**0**
one-row ribbon	31	218	48	85

So the failure sits where Rick's Remark II.11 says it does.

Independent computation. $X_{\lambda/\mu}$ was built from KLO's App. A tableau rule, not from the word model. I certified $M \cong \operatorname{Ind} V$ by dimension together with BFS generation, checked every extension for H_n-linearity, and computed t by trace-form rank. Prediction (Cor. C): $t = \prod_{\text{maximal ribbon runs}} 2^{r-1}$.

- $n \leq 5$, all disconnected shapes: 92 shapes, 0 mismatches; 2 hypothesis shapes, 90 controls all split.
- $n = 6, 7$, shapes with at least one non-ribbon component: 63 shapes, 0 mismatches; 53 hypothesis shapes all have $t = 1$; 10 controls (for example $(2,2) \star (1) \star (1)$) all split. **32** shapes at $n = 7$ were skipped by my size cap. That is a coverage gap.
- Cor. II.7 (multiplicity-free top of size 2^{m-1}): 232/232.

4 Cited results (brief Q5)

I read the KLO source (arXiv:2310.20571, e-print **v2, 2025-01-20**) directly and reconstructed the shared counter `\newtheorem{theorem}{Theorem}[section]`.

cited	in KLO	status
Lemma 5.1	Huang decomposition, cited to Huang Thm 3.3	located; Rick re-proves it (Rem. II.10), and I checked the re-proof: **proved**
Lemma 5.3(1)	read of $T_{\lambda/\mu}$	located; used only in II.6, which I checked computationally (82/82)
Lemma 5.4(1)	projective cover $P_{\alpha_{\text{proj}}}$	**located, proved**
Thm 5.5	$M_P \cong M_Q \iff$ same shape	**located, proved**; KLO's "only if" infers the shape from per-component row and column lengths. I checked that inference on all basic skew shapes with $n \leq 8$: 0 collisions (2659 shapes at $n = 8$). Corroboration only
Prop. 7.3(1)	connected $\Rightarrow$ indecomposable	**located, proved**
Rem. A.1(2)	ribbon modules projective	**located**, a remark with no separate proof; consistent with 90/90 all-ribbon splits
Conj. 7.5	§7.2	matches Thm B; "contains a disconnected ribbon" is defined on consecutive connected components

5 Owner gate

This is a claim about absence. KLO v2 (Jan 2025) still states Conj. 7.5 as open. Semantic Scholar lists **6** citers, not the 12 Rick reports; S2 drops edges, so it fails toward "open". I grepped the e-prints of 5 citers, Defant–Searles 2404.04961, and 2601.13324 for "disconnected ribbon", "Conjecture 7.5" and the KLO label, with "indecomposab" as a positive control. None addresses the conjecture. Remaining failure modes: a proof that does not cite KLO, an unlinked paper, unpublished KLO-school work. I agree with *not found*, and the public-claim gate should stay.

6 Requested changes

S1. State Lemma II.8 for "C_h has at least two rows". Then say that "not a ribbon" is used only in §13 Step 5, which promotes Remark II.11 into the structure of the proof.

S2. Name the instrument that returned 12 citers (S2 returns 6), and list them.

S3. Consider stating Thm B as the dichotomy "$X_{\lambda/\mu}$ is indecomposable iff λ/μ contains no disconnected ribbon" (with $m = 1$ and the "only if" from KLO Prop. 7.3).

N1. §5, "eigenspaces multiply": "commuting" is not needed.

N2. Lemma II.5(b) names $\widehat{L}$ before (a)'s second inclusion has shown $\widehat{L}' = \widehat{L}$. Add a forward reference.

N3. §16: say which split the Wake 289 sweep used.

N4. Rem. II.12's (G*) counterexamples were *not* checked by me, and the endorsement does not cover that remark.

7 Connections

Rick's §5 gives, without stating it, $\dim \operatorname{End} M_{I,J} = 1 + \sum_{k \geq 1} \dim \operatorname{Hom}(C_I, L_k) \dim \operatorname{Hom}(C_J, R_k)$, a recursive answer to the dimension half of Q72(ii). Separately, the trace-form rank used here is independent of the top, which is what made it a fair second instrument for a proof that runs on the top.